



UNIT 02: QUANTUM COMPUTING FUNDAMENTALS

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INTRODUCTION

What if a single chip, smaller than your thumbnail, could solve a problem that would take the world's most powerful supercomputer longer than the age of the universe?

In December 2024, Google unveiled Willow — a quantum processor with 105 qubits that completed a benchmark computation in under five minutes. The same task, performed on today's fastest classical supercomputer, would take an estimated 10 septillion years (10^{25}) — a number so vast it dwarfs the 13.8-billion-year age of the universe. This was not science fiction; it was a laboratory result, published in *Nature* and independently verified.

The secret behind this staggering power lies in three phenomena that defy everyday intuition: **superposition** (a quantum bit existing as 0 and 1 simultaneously), **entanglement** (particles correlated in ways that have no classical explanation), and **interference** (amplitudes combining to amplify correct answers and cancel wrong ones). These are not abstract curiosities — they are the working principles of a new kind of computation.

This unit equips you with the mathematical and conceptual tools to understand exactly how these quantum phenomena enable computational power beyond classical reach. We explore qubits and their mathematical representation, quantum gates that manipulate qubit states, and the measurement process that extracts classical information from quantum systems. The unit progresses from single-qubit concepts to multi-qubit systems, culminating in entanglement and teleportation.

Before studying this unit, you should have:

- Completion of Unit 1: Foundations of Classical Machine Learning
- Basic linear algebra: vectors, matrices, matrix multiplication
- Complex numbers: magnitude, phase, conjugate
- Probability concepts: discrete distributions, expected values



LEARNING OBJECTIVES

Upon successful completion of this unit, learners will be able to:

1. Analyze single-qubit quantum gates including Pauli gates (X, Y, Z), the Hadamard gate, and rotation gates, comparing their matrix representations and decomposing arbitrary single-qubit operations
2. Apply multi-qubit gates such as CNOT and controlled-U operations to construct and analyze Bell states, demonstrating quantum entanglement and the no-cloning theorem
3. Evaluate quantum measurement strategies by distinguishing between projective measurements and POVMs, and justify the choice of measurement basis for different quantum tasks
4. Design simple quantum circuits by combining single- and multi-qubit gates to prepare specific quantum states and verify circuit correctness
5. Analyze quantum state representations using Dirac notation, the Bloch sphere, and density matrices to interpret superposition and entanglement phenomena

2.1 QUANTUM MECHANICS BASICS, QUANTUM STATES

2.1.1 FROM CLASSICAL BITS TO QUANTUM BITS

In classical computing, information is encoded in **bits**, the fundamental unit of binary information. Each classical bit exists in one of exactly two states: 0 or 1. This binary representation forms the basis of all digital computers, where complex information is encoded as sequences of bits. For instance, a set of three classical bits can represent $2^3 = 8$ distinct states, ranging from 000 to 111.

While classical bits have enabled remarkable computational achievements, they are fundamentally limited by their binary nature. Each bit must be definitively in state 0 or



state 1 at any given moment. This limitation becomes significant when addressing certain computational problems that exhibit exponential complexity.

Quantum computing introduces a revolutionary concept: the **quantum bit** or **qubit**. Unlike classical bits, qubits exploit principles of quantum mechanics to exist in a fundamentally different state space. A qubit can exist not only in states $|0\rangle$ or $|1\rangle$, but also in a **superposition** of both states simultaneously. This property, combined with quantum entanglement and interference, provides the foundation for quantum computational advantage.

The transition from classical to quantum computing constitutes a paradigm shift: where a classical computer with n bits represents one of 2^n states at a time, a quantum computer with n qubits can represent a superposition of all 2^n basis states, enabling fundamentally new computational approaches through quantum interference.

2.1.2 MATHEMATICAL REPRESENTATION OF QUBITS

A single-qubit state is mathematically represented as a two-dimensional complex vector with unit length. Using Dirac notation, we write:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers called **probability amplitudes**, and $|0\rangle$ and $|1\rangle$ are the **computational basis states** defined as:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The amplitudes must satisfy the **normalization constraint**:

$$|\alpha|^2 + |\beta|^2 = 1$$

This constraint ensures that probabilities sum to unity. The physical interpretation is that $|\alpha|^2$ represents the probability of measuring the qubit in state $|0\rangle$, while $|\beta|^2$ represents the probability of measuring it in state $|1\rangle$.

Since α and β are complex numbers, we can write them as:

$$\alpha = |\alpha|e^{i\phi_\alpha}, \beta = |\beta|e^{i\phi_\beta}$$



The relative phase between amplitudes $(\phi_\beta - \phi_\alpha)$ has physical significance and affects quantum interference, while the global phase can be ignored.

More generally, a qubit state can be parameterized using real parameters:

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right)$$

where $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$, and γ is an ignorable global phase.

2.1.3 SUPERPOSITION PRINCIPLE

Quantum superposition is the phenomenon where a qubit exists in a coherent combination of basis states rather than being definitively in one state or the other. When we write:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with both $\alpha \neq 0$ and $\beta \neq 0$, the qubit is simultaneously in states $|0\rangle$ and $|1\rangle$. This is not simply a statement of ignorance about which state the qubit occupies—it is a fundamentally different kind of state that has no classical analog.

Superposition differs crucially from classical probability. In classical probability, if we say a coin has a 50% chance of being heads, the coin is definitely one or the other—we just don't know which. In quantum superposition, the qubit genuinely exists in both states simultaneously until measured.

The power of superposition becomes apparent in computation. Consider a quantum operation applied to a superposition:

$$U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle$$

The operation is effectively applied to both basis states due to linearity. With n qubits in superposition, a single quantum operation acts on all 2^n amplitudes simultaneously. This quantum parallelism is a key resource for quantum algorithms, though it yields useful speedups only when combined with interference.

However, extracting useful information from superposition requires careful algorithm design, as measurement collapses the superposition to a single classical outcome.

Quantum algorithms must use interference to amplify correct answers while canceling incorrect ones.

2.1.4 THE BLOCH SPHERE

The **Bloch sphere** provides a powerful geometric visualization of single-qubit pure states. Since a qubit state has two complex parameters satisfying one normalization constraint, the effective parameter space is three-dimensional after accounting for the irrelevant global phase. This maps naturally onto a sphere.

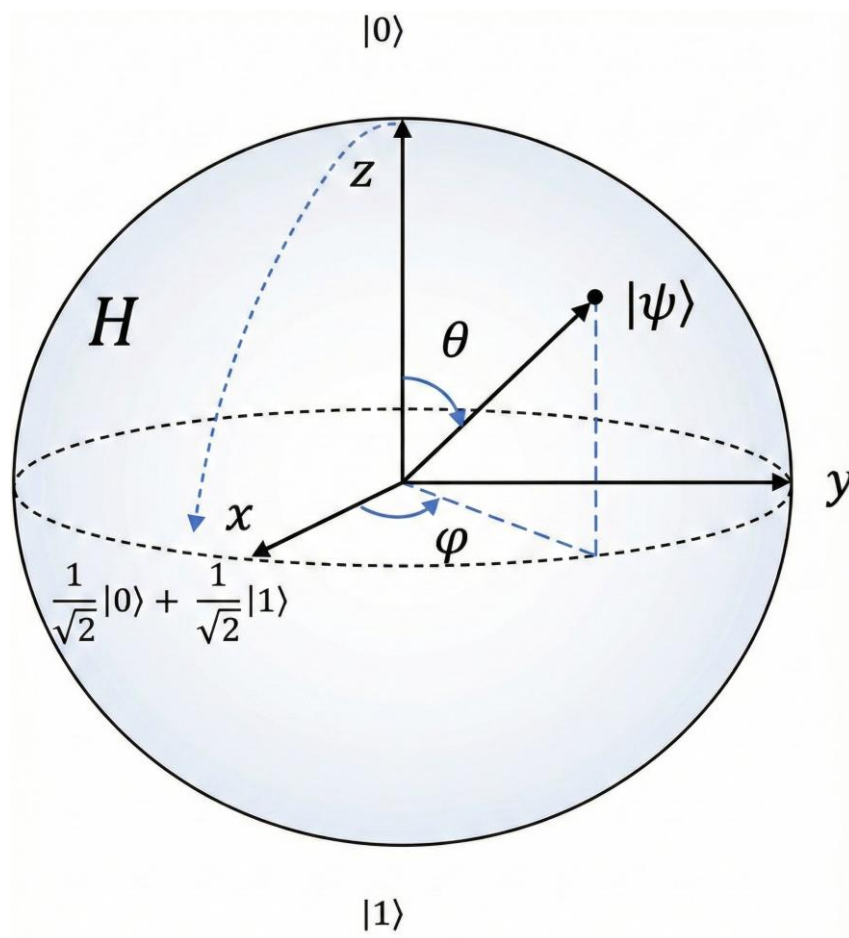


Figure 2.1: Bloch Sphere Representation Showing the Hadamard Gate Transformation from $|0\rangle$ to the Equal Superposition State



Any pure single-qubit state can be represented as a point on the surface of the Bloch sphere using spherical coordinates:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where:

- θ is the **polar angle** (angle from the positive z-axis), with $\theta \in [0, \pi]$
- ϕ is the **azimuthal angle** (angle in the xy-plane from the positive x-axis), with $\phi \in [0, 2\pi)$

Key states on the Bloch sphere include:

State	Location	Coordinates
$ 0\rangle$	North pole	$\theta = 0$
$ 1\rangle$	South pole	$\theta = \pi$
$ +\rangle = (0\rangle + 1\rangle)/\sqrt{2}$	Positive x-axis	$\theta = \pi/2, \phi = 0$
$ -\rangle = (0\rangle - 1\rangle)/\sqrt{2}$	Negative x-axis	$\theta = \pi/2, \phi = \pi$
$ +i\rangle = (0\rangle + i 1\rangle)/\sqrt{2}$	Positive y-axis	$\theta = \pi/2, \phi = \pi/2$
$ -i\rangle = (0\rangle - i 1\rangle)/\sqrt{2}$	Negative y-axis	$\theta = \pi/2, \phi = 3\pi/2$

Table 2.1: Key Quantum States on the Bloch Sphere.

The Bloch sphere representation allows quantum gates to be visualized as rotations: the Pauli-X gate rotates by π around the x-axis, the Hadamard gate rotates by π around an axis halfway between x and z, and parameterized rotation gates rotate by their parameter angle about the corresponding axis.

2.1.5 DIRAC NOTATION

Dirac notation (bra-ket notation) provides a compact and elegant mathematical framework for quantum mechanics. This notation was introduced by physicist Paul Dirac and is universally used in quantum computing.

Ket notation $|\psi\rangle$ represents a quantum state as a column vector:



$$|\psi\rangle = (\alpha \ \beta)$$

Bra notation $\langle\psi|$ represents the conjugate transpose (Hermitian adjoint) of the ket:

$$\langle\psi| = |\psi\rangle^\dagger = (\alpha^*, \beta^*)$$

The **inner product** (or bracket) $\langle\phi|\psi\rangle$ produces a scalar:

$$\langle\phi|\psi\rangle = \phi_1^* \psi_1 + \phi_2^* \psi_2 \in \mathbb{C}$$

The inner product has important properties:

- $\langle\psi|\psi\rangle = \|\psi\|^2$ (norm squared)
- $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$ (conjugate symmetry)
- If $\langle\phi|\psi\rangle = 0$, the states are orthogonal

For the computational basis, orthonormality holds:

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \langle 0|1\rangle = \langle 1|0\rangle = 0$$

The **outer product** $|\phi\rangle\langle\psi|$ creates a matrix (operator):

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 00 \end{pmatrix}$$

The identity operator can be expressed using the completeness relation:

$$I = |0\rangle\langle 0| + |1\rangle\langle 1|$$

This notation extends naturally to operators. For an operator A acting on state $|\psi\rangle$, we write $A|\psi\rangle$, and the expected value is $\langle\psi|A|\psi\rangle$.

2.1.6 MULTI-QUBIT STATES

When we have multiple qubits, their combined state is described using the **tensor product** (Kronecker product). For two single-qubit states:

$$|a\rangle \otimes |b\rangle = (a_1 \ a_2) \otimes (b_1 \ b_2) = (a_1 b_1 \ a_1 b_2 \ a_2 b_1 \ a_2 b_2)$$

For convenience, we often write $|a\rangle \otimes |b\rangle$ as $|a\rangle|b\rangle$, $|a, b\rangle$, or $|ab\rangle$.

The **two-qubit computational basis** consists of four states:

$$|00\rangle = (1 \ 0 \ 0 \ 0), |01\rangle = (0 \ 1 \ 0 \ 0), |10\rangle = (0 \ 0 \ 1 \ 0), |11\rangle = (0 \ 0 \ 0 \ 1)$$

For an **N-qubit system**, the state space has dimension 2^N :

$$|\psi\rangle = \sum_{i=0}^{2^N-1} c_i |i\rangle \in \mathbb{C}^{2^N}$$



where the coefficients satisfy $\sum_i |c_i|^2 = 1$, and $|i\rangle$ represents the computational basis state corresponding to the binary representation of i .

This exponential growth of state space is both the promise and challenge of quantum computing. For 50 qubits, the state vector has approximately 10^{15} complex amplitudes—more than can be stored in any classical computer. However, when the number of nonzero entries in c_i is larger than one (indicating superposition), we say the state is in superposition.

Not all multi-qubit states can be written as tensor products of single-qubit states. States that cannot be so factored are called **entangled states**, which we explore further in Section 2.2.

2.2 QUANTUM GATES, SUPERPOSITION AND ENTANGLEMENT

2.2.1 CLASSICAL LOGIC GATES REVIEW

Classical digital logic circuits process bits through **logic gates** (NOT, AND, OR, XOR) that implement Boolean functions. Classical gates have **universality** (NAND or NOR alone can implement any Boolean function), but most are **irreversible** — the AND gate maps three input combinations to the same output, making it impossible to determine the input from the output. Quantum gates, by contrast, must be reversible due to the unitary nature of quantum evolution.

A	B	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0



Table 2.2: Truth Table for Classical Logic Gates.

2.2.2 UNITARY OPERATIONS

Quantum gates are mathematical operations that transform qubit states. Unlike classical gates, all quantum gates must be **unitary transformations**.

A matrix U is **unitary** if:

$$U^\dagger U = U U^\dagger = I$$

where $U^\dagger$ is the conjugate transpose of U and I is the identity matrix.

Unitary operations have several important properties:

1. **Reversibility:** Since $U^{-1} = U^\dagger$, every quantum gate has an inverse
2. **Norm preservation:** $\|U|\psi\rangle\| = \||\psi\rangle\|$, ensuring $\langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle = 1$
3. **Inner product preservation:** $\langle U\phi|U\psi\rangle = \langle\phi|\psi\rangle$

When a unitary U acts on a state $|a\rangle$, the evolution is:

$$|\hat{a}\rangle = U|a\rangle = \hat{a}_1|0\rangle + \hat{a}_2|1\rangle$$

The normalization $|\hat{a}_1|^2 + |\hat{a}_2|^2 = \langle\hat{a}|\hat{a}\rangle = \langle a|U^\dagger U|a\rangle = 1$ is preserved if and only if U is unitary.

2.2.3 PAULI GATES

The **Pauli gates** are fundamental single-qubit operations corresponding to the Pauli matrices. They form a basis for single-qubit operations and appear throughout quantum computing.

Pauli-X (NOT gate): Performs a bit flip

$$X = (01 \ 10)$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Action on basis states:



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- $X|0\rangle = |1\rangle$

- $X|1\rangle = |0\rangle$

$$[|0\rangle = (1, 0)^T, \quad |1\rangle = (0, 1)^T]$$

On the Bloch sphere, X is a rotation by π around the x-axis.

Pauli-Y: Combines bit flip with phase flip

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Action on basis states:

- $Y|0\rangle = i|1\rangle$

- $Y|1\rangle = -i|0\rangle$

Pauli-Z (Phase gate): Performs a phase flip

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Action on basis states:

- $Z|0\rangle = |0\rangle$

- $Z|1\rangle = -|1\rangle$

The Z gate can also be written using outer products:

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

All Pauli matrices satisfy $X^2 = Y^2 = Z^2 = I$ (self-inverse) and the product relation $XY = iZ$ (and cyclic permutations: $YZ = iX$, $ZX = iY$). Distinct Pauli matrices also anticommute: $\{X, Y\} = XY + YX = 0$.



Summarization of Quantum Gates

Abbreviation, matrix form, and circuit representation

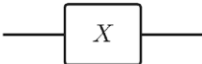
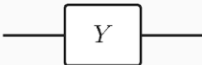
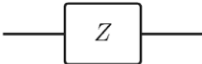
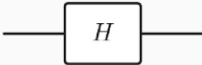
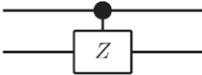
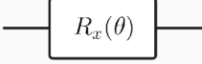
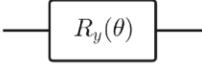
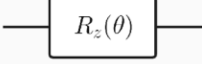
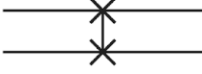
Quantum Gate	Matrix Form	Circuit Representation
Pauli-X (X)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	
Pauli-Y (Y)	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	
Pauli-Z (Z)	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	
Hadamard (H)	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	
Controlled-Z (CZ)	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	
$R_X(\theta)$	$\begin{pmatrix} \cos(\frac{\theta}{2}) & -i \sin(\frac{\theta}{2}) \\ -i \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$	
$R_Y(\theta)$	$\begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$	
$R_Z(\theta)$	$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$	
SWAP	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	
Controlled-NOT (CNOT)	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	

Figure 2.2: Summarization of Quantum Gates Showing Abbreviation, Matrix Form, and Circuit Representation



2.2.4 HADAMARD GATE

The **Hadamard gate** is one of the most important quantum gates, creating superposition from basis states.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Action on basis states:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Key property: The Hadamard gate is **self-inverse**:

$$H^2 = H \cdot H = I$$

This means applying H twice returns the qubit to its original state.

On the Bloch sphere, the Hadamard gate rotates by π around an axis that lies at 45° between the x and z axes. It maps the north pole ($|0\rangle$) to the positive x-axis ($|+\rangle$), and the south pole ($|1\rangle$) to the negative x-axis ($|-\rangle$).

The Hadamard gate is essential for:

- Creating superposition at the start of quantum algorithms
- Changing between computational and Hadamard bases
- Implementing quantum interference in algorithms like Grover's search

2.2.5 ROTATION GATES

Rotation gates are parameterized single-qubit gates that rotate the state on the Bloch sphere by a specified angle around a specified axis.

$R_X(\theta)$ - **Rotation about X-axis:**

$$R_X(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

$R_Y(\theta)$ - **Rotation about Y-axis:**

$$R_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$



$R_Z(\theta)$ - **Rotation about Z-axis:**

$$R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

An important theorem states that any single-qubit unitary operation can be decomposed into a sequence of rotations:

$$U = R_Z(\alpha)R_Y(\beta)R_Z(\gamma)$$

Here $\alpha, \beta, \gamma \in [0, 2\pi)$ are Euler rotation angles (not to be confused with the probability amplitudes from Section 2.1.2), up to a global phase. This ZYZ decomposition is crucial for implementing arbitrary single-qubit operations on hardware that natively supports only certain rotation gates.

2.2.6 CNOT GATE

The **Controlled-NOT (CNOT)** gate is the most important two-qubit gate. It operates on a **control qubit** and a **target qubit**.

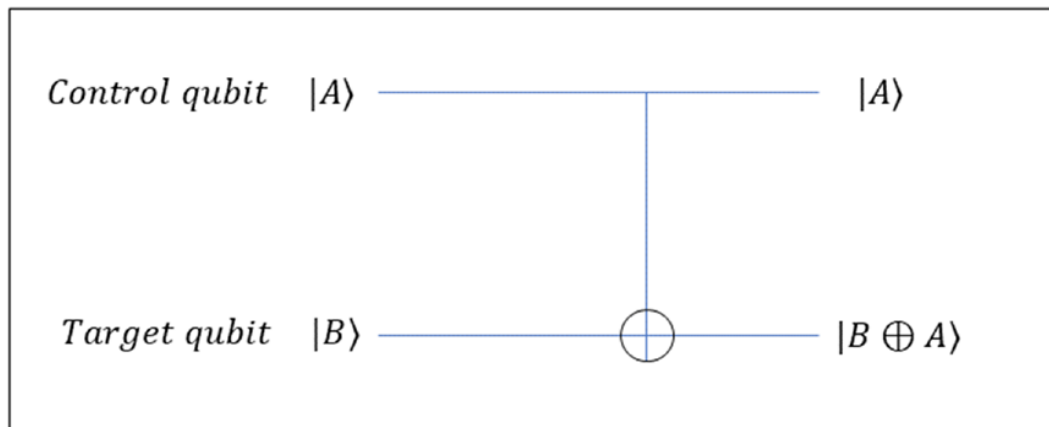


Figure 2.3: CNOT Gate Logical Circuit Showing Control Qubit, Target Qubit, and Output States

The CNOT gate's behavior:

- If the control qubit is $|0\rangle$, the target qubit remains unchanged
- If the control qubit is $|1\rangle$, the target qubit is flipped (NOT applied)

Truth table:



Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

Table 2.3: Truth Table for the CNOT Gate.

The CNOT operation can be expressed as:

$$CNOT: |A, B\rangle \rightarrow |A, B \oplus A\rangle$$

where $\oplus$ denotes addition modulo 2 (XOR).

The matrix representation (4×4):

$$U_{CNOT} = (1000 \ 0100 \ 0001 \ 0010)$$

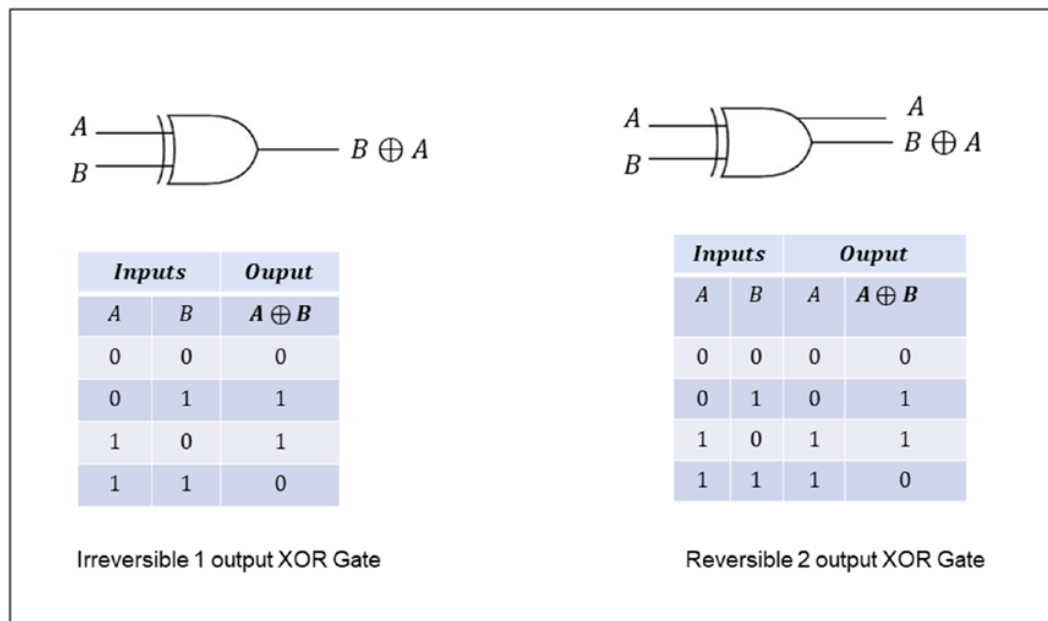


Figure 2.4: Comparison of Irreversible and Reversible XOR Gates with Truth Tables

Unlike the classical XOR gate which is irreversible, the CNOT gate is reversible—given the output, we can uniquely determine the input. This is because the CNOT gate is its own inverse: $CNOT^2 = I$.

The CNOT gate is crucial for:



- Creating entanglement (combined with H gate)
- Implementing classical reversible logic
- Building more complex multi-qubit gates

2.2.7 OTHER MULTI-QUBIT GATES

Beyond CNOT, several other multi-qubit gates are commonly used.

Controlled-Z (CZ) Gate: Applies a phase flip when both qubits are $|1\rangle$

$$CZ = (1000 \ 0100 \ 0010 \ 000 - 1)$$

SWAP Gate: Exchanges the states of two qubits

$$SWAP = (1000 \ 0010 \ 0100 \ 0001)$$

Controlled-U Gate: A generalization where any unitary U is applied to target qubit(s) conditioned on the control qubit being $|1\rangle$. The CNOT is a special case where $U = X$.

Universal Gate Sets: While many single and multi-qubit gates exist, it is sufficient to use a **universal set of gates** to construct any unitary operation. A commonly used universal gate set includes:

- Single-qubit rotations: $R_X(\theta)$, $R_Y(\theta)$, $R_Z(\theta)$
- Two-qubit gate: CNOT

The **Solovay-Kitaev theorem** states that any unitary operator can be approximated to arbitrary precision $\varepsilon > 0$ by a finite sequence of gates from a universal gate set, with the number of gates scaling as $O(\log^c(1/\varepsilon))$ where $c \approx 3.97$.

2.2.8 QUANTUM CIRCUITS

Quantum circuits are the standard way to represent quantum algorithms. A quantum circuit consists of:

- **Wires:** Horizontal lines representing qubits, with time flowing left to right
- **Gates:** Boxes or symbols placed on wires representing quantum operations
- **Measurements:** Meter symbols indicating where measurements occur

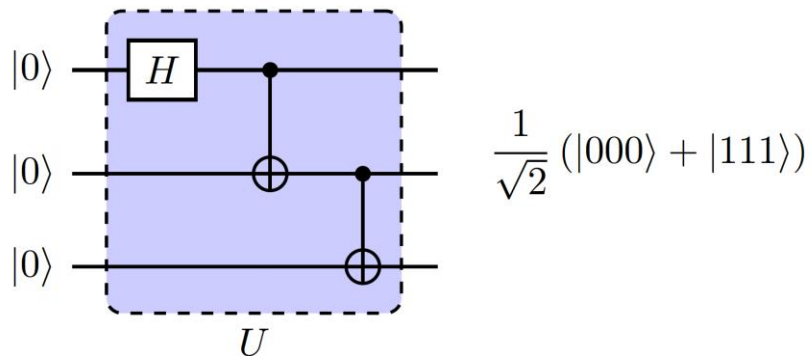


Figure 2.5: Decomposition of the Multi-qubit Circuit for the 3-qubit GHZ State

In circuit diagrams:

- Single-qubit gates are shown as boxes on one wire
- CNOT is shown with a dot (control) connected vertically to $\oplus$ (target)
- Multi-qubit gates span multiple wires

When multiple gates appear at the same horizontal position, they are applied simultaneously (if acting on different qubits) or in sequence from left to right.

The circuit model serves as a foundational framework for quantum computation because:

1. It provides a standardized graphical language for quantum algorithms
2. It enables clear visualization of computational flow
3. It allows quantum operations to be decomposed into a pre-defined gate set
4. It ensures compatibility across different quantum hardware architectures

2.2.9 BELL STATES

Bell states (or EPR pairs) are the four maximally entangled two-qubit states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$



$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

These states exhibit perfect correlation: measuring one qubit immediately determines the other's state. For $|\Phi^+\rangle$:

- If qubit A measures 0, qubit B will certainly measure 0
- If qubit A measures 1, qubit B will certainly measure 1

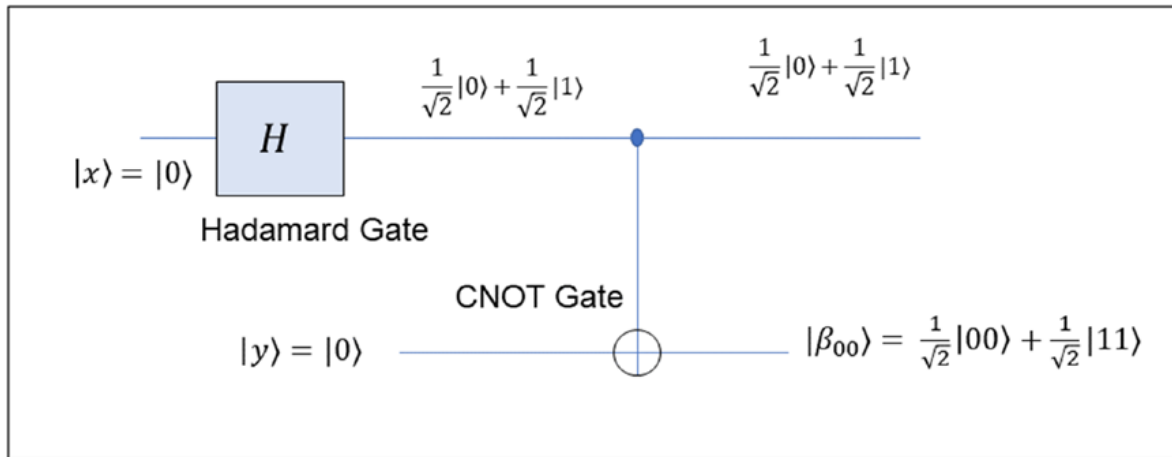


Figure 2.6: Bell State Implementation Using Hadamard and CNOT Gates

Creating Bell States: Starting with $|00\rangle$:

1. Apply Hadamard to first qubit: $(1/\sqrt{2})(|0\rangle + |1\rangle)|0\rangle$
2. Apply CNOT: $(1/\sqrt{2})(|00\rangle + |11\rangle) = |\Phi^+\rangle$

The generalized Bell state formula:

$$|\beta_{xy}\rangle = \frac{1}{\sqrt{2}}|0, y\rangle + \frac{(-1)^x}{\sqrt{2}}|1, \bar{y}\rangle$$

Here $x, y \in \{0,1\}$ select which Bell state is produced, $\bar{y}$ denotes the complement of y , and $(-1)^x$ determines the relative phase. Bell states form an orthonormal basis for the two-qubit Hilbert space and are fundamental resources for quantum communication and computation.



2.2.10 PROPERTIES OF ENTANGLEMENT

Quantum entanglement is a uniquely quantum phenomenon where the quantum state of a multi-qubit system cannot be described as a product of individual qubit states.

Formally, an N-qubit state $|\psi\rangle \in \mathcal{C}^{2^N}$ is **entangled** if it cannot be expressed as:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

for any partition into subsystems A and B. If it can be so expressed, it is called **separable**.

Example of a separable state:

$$|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

Example of an entangled state (Bell state):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This cannot be factored into separate states for each qubit.

Properties of entangled states:

1. **Non-locality:** Measurements on entangled qubits exhibit correlations that cannot be explained by classical physics (Bell's theorem)
2. **Monogamy:** Maximal entanglement between two qubits limits their entanglement with other qubits
3. **Resource:** Entanglement is a resource for quantum computation, communication, and cryptography

2.2.11 THE NO-CLONING THEOREM

The **no-cloning theorem** states that it is impossible to create an exact copy of an arbitrary unknown quantum state. This is a fundamental result that distinguishes quantum from classical information.

Theorem: There is no quantum operation that can perform the transformation $|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle$ for all states $|\psi\rangle$.

Proof sketch: Suppose we try to copy state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ using a CNOT gate:



Input: $|\psi\rangle|0\rangle = (\alpha|0\rangle + \beta|1\rangle)|0\rangle = \alpha|00\rangle + \beta|10\rangle$

After CNOT: $\alpha|00\rangle + \beta|11\rangle$

If copying succeeded, we would need: $(\alpha|0\rangle + \beta|1\rangle)(\alpha|0\rangle + \beta|1\rangle) = \alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle$

Comparing: $\alpha|00\rangle + \beta|11\rangle \neq \alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|10\rangle + \beta^2|11\rangle$

These are equal only if $\alpha\beta = 0$, meaning the state is $|0\rangle$ or $|1\rangle$ (not a superposition).

Implications:

- Quantum states cannot be backed up or copied
- This enables quantum cryptography (eavesdropping disturbs the state)
- Classical information can still be transmitted (measure and send the result)
- The no-cloning theorem does not prevent quantum teleportation (where the original is destroyed)

2.2.12 QUANTUM TELEPORTATION

Quantum teleportation is a protocol that transmits a quantum state from one location to another using entanglement and classical communication.

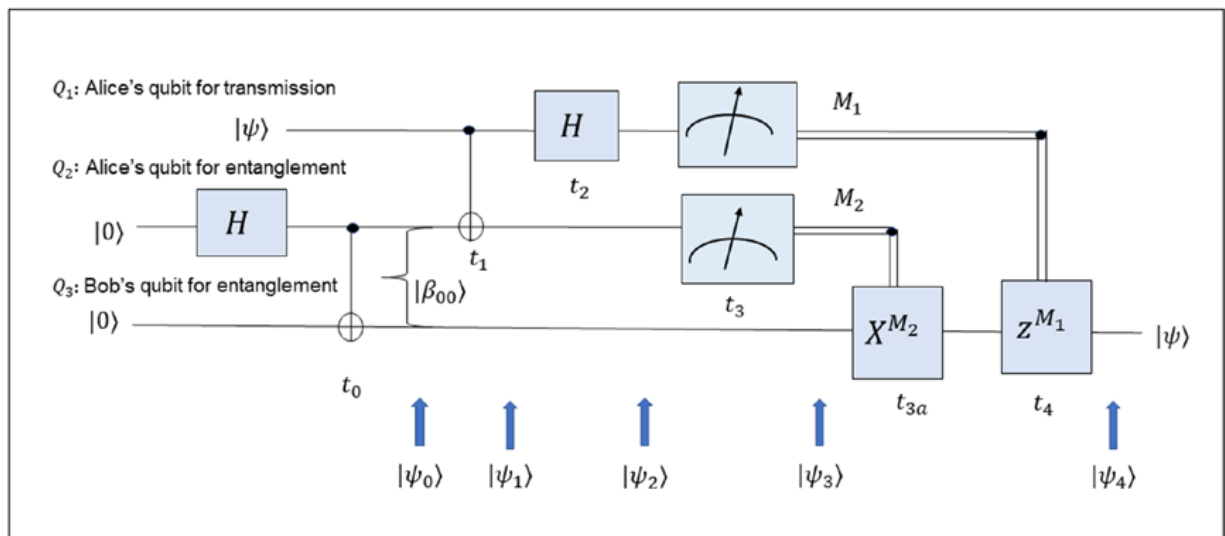


Figure 2.7: Quantum Teleportation Circuit Showing Alice and Bob's Operations with Classical Communication



Protocol: Alice wants to send state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob.

Setup: Alice and Bob share an entangled Bell state $|\beta_{00}\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle)$, with Alice holding qubit Q_2 and Bob holding qubit Q_3 .

Steps:

1. **Initial state:** $|\psi\rangle \otimes |\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (1/\sqrt{2})(|00\rangle + |11\rangle)$
2. **Alice applies CNOT(Q_1, Q_2):** Entangles her qubits
3. **Alice applies H(Q_1):** Creates interference
4. **Alice measures Q_1 and Q_2 :** Gets classical bits M_1, M_2
5. **Alice sends M_1, M_2 to Bob:** Classical communication
6. **Bob applies corrections:** $Z^{M_1}X^{M_2}$ to Q_3

After Bob's corrections, his qubit Q_3 is in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

Key points:

- The original state $|\psi\rangle$ is destroyed at Alice's end (no violation of no-cloning)
- Classical communication is required (no faster-than-light information transfer)
- The quantum state is "teleported," not physically moved
- Teleportation requires pre-shared entanglement

2.3 QUANTUM MEASUREMENTS

2.3.1 THE MEASUREMENT PROBLEM

Quantum measurement is the process of extracting classical information from a quantum system. Unlike classical measurement, quantum measurement fundamentally disturbs the system being measured.

When we measure a qubit in superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the measurement:

1. Returns a classical outcome (0 or 1)
2. **Collapses** the superposition to the corresponding basis state
3. Produces probabilistic outcomes according to Born's rule



After measurement, the quantum state is irrevocably changed. If we measure 0, the post-measurement state is $|0\rangle$ regardless of the original amplitudes. The superposition information is lost.

This probabilistic, destructive nature of measurement is what makes quantum computing both challenging and powerful. Algorithms must be designed so that:

- Useful information can be extracted through measurement
- Interference amplifies correct answers before measurement
- Multiple runs allow statistical estimation of results

2.3.2 PROJECTIVE MEASUREMENTS

The standard quantum measurement formalism is **projective measurement** (also called von Neumann measurement).

A projective measurement is described by a Hermitian operator A with spectral decomposition:

$$A = \sum_i \lambda_i |v_i\rangle\langle v_i|$$

Here λ_i are the real eigenvalues (possible measurement outcomes) and $|v_i\rangle$ are the corresponding orthonormal eigenvectors.

When measuring state $|\Phi\rangle$, the probability of obtaining eigenvalue λ_i is given by **Born's rule**:

$$Pr(\lambda_i) = |\langle v_i | \Phi \rangle|^2$$

After obtaining outcome λ_i , the state collapses to the corresponding eigenvector $|v_i\rangle$.

For measurement in the **computational basis**, we use projectors:

$$\Pi_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \Pi_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

- $Pr(0) = |\langle 0 | \psi \rangle|^2 = |\alpha|^2$
- $Pr(1) = |\langle 1 | \psi \rangle|^2 = |\beta|^2$

These projectors satisfy: $\Pi_0^2 = \Pi_0$, $\Pi_1^2 = \Pi_1$, and $\Pi_0 + \Pi_1 = I$.



2.3.3 MEASUREMENT IN DIFFERENT BASES

Measurements need not be performed in the computational basis. The **Hadamard basis** (or X-basis) consists of:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

We can express computational basis states in the Hadamard basis:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

To measure in the Hadamard basis using a device that only measures in the computational basis:

1. Apply the Hadamard gate H to the qubit
2. Measure in the computational basis

If measuring $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the Hadamard basis, we will observe:

- $|+\rangle$ with probability $|\alpha + \beta|^2/2$
- $|-\rangle$ with probability $|\alpha - \beta|^2/2$

The choice of measurement basis affects which information can be extracted from a quantum state, a principle exploited in quantum cryptography.

2.3.4 POVM MEASUREMENTS

Positive Operator-Valued Measures (POVM) generalize projective measurements. A POVM is described by a collection of positive operators $\{E_i\}$ satisfying:

$$E_i \geq 0 \text{ and } \sum_i E_i = I$$

The probability of outcome i when measuring state $|\psi\rangle$:

$$Pr(i) = \langle\psi|E_i|\psi\rangle$$

Each E_i is a positive semi-definite operator, and the collection satisfies $\sum_i E_i = I$.



For density matrices: $Pr(i) = Tr(\rho E_i)$

The key difference from projective measurements: POVM elements do not have to be orthogonal. This allows for measurements that distinguish between non-orthogonal states with some probability of error.

Projective measurement is a special case of POVM where $E_i = \Pi_i^\dagger \Pi_i = \Pi_i$ (since projectors are Hermitian and idempotent) and the projectors are mutually orthogonal: $\Pi_i \Pi_j = \delta_{ij} \Pi_j$.

2.3.5 QUANTUM CIRCUIT REPRESENTATION

[Image: Quantum Circuit with Measurement]

Figure 2.8: Quantum Circuit Diagram Showing Unitary Gate Followed by Measurement

In quantum circuits, measurements are depicted using a **meter symbol**. The measurement operation:

1. Accepts a quantum input (the qubit to be measured)
2. Produces a classical output (the measurement result)

After measurement, the wire typically becomes a **double line** indicating it carries classical information. This classical information can control later quantum operations (classical control), as seen in quantum teleportation and error correction protocols.

2.3.6 PURE VS MIXED STATES

So far, we have described quantum states as **pure states** using state vectors $|\psi\rangle$. Pure states represent the maximum possible knowledge about a quantum system—the system is in a definite quantum state.

However, in practice, we often encounter situations where:

- The quantum system interacts with an environment
- We have incomplete knowledge about which pure state was prepared
- We're describing part of an entangled system



In these cases, we need **mixed states**, which represent statistical ensembles of pure states. Mixed states are described using the **density operator** (or density matrix) formalism.

2.3.7 DENSITY MATRIX REPRESENTATION

An N-qubit **density operator** $\rho \in \mathcal{C}^{2^N \times 2^N}$ represents a mixture of m quantum pure states $|\psi_i\rangle$ with probabilities p_i :

$$\rho = \sum_{i=1}^m p_i |\psi_i\rangle\langle\psi_i|$$

Here p_i are classical probabilities satisfying $\sum_i p_i = 1$, and $|\psi_i\rangle\langle\psi_i|$ is the projector onto pure state $|\psi_i\rangle$.

Properties of density matrices:

1. **Hermitian:** $\rho = \rho^\dagger$
2. **Positive semi-definite:** $\langle v|\rho|v\rangle \geq 0$ for all $|v\rangle$
3. **Trace one:** $Tr(\rho) = 1$

Purity criterion:

- Pure state ($m = 1$): $Tr(\rho^2) = 1$
- Mixed state ($m > 1$): $Tr(\rho^2) < 1$

Example 1 - Pure state $|\psi\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$:

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$Tr(\rho^2) = Tr(\rho) = 1 \checkmark \text{ (confirms pure state)}$$

Example 2 - Mixed state (equal mixture of $|0\rangle$ and $|1\rangle$):

$$\rho = 0.5|0\rangle\langle 0| + 0.5|1\rangle\langle 1| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Tr(\rho^2) = 0.5 < 1 \checkmark \text{ (confirms mixed state)}$$

Note that while the pure superposition $(1/\sqrt{2})(|0\rangle + |1\rangle)$ has off-diagonal elements (coherence), the mixed state has zero off-diagonal elements, indicating no quantum coherence.



2.3.8 EVOLUTION AND MEASUREMENT WITH DENSITY MATRICES

Unitary evolution of density matrices:

$$\hat{\rho} = U\rho U^\dagger$$

This follows from $|\hat{\psi}\rangle = U|\psi\rangle$, so $\hat{\rho} = |\hat{\psi}\rangle\langle\hat{\psi}| = U|\psi\rangle\langle\psi|U^\dagger = U\rho U^\dagger$.

Measurement probabilities with density matrices:

$$Pr(\lambda_i) = Tr(\rho|v_i\rangle\langle v_i|) = Tr(\rho\Pi_i)$$

The density matrix formalism is essential for:

- Describing open quantum systems interacting with environments
- Analyzing quantum decoherence and noise
- Partial traces when studying subsystems of entangled states

2.4 SUMMARY

This unit covered the fundamental concepts of quantum computing necessary for understanding quantum machine learning:

Quantum States: Qubits extend classical bits by existing in superpositions $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, visualized on the Bloch sphere and manipulated using Dirac notation. Multi-qubit states are described via tensor products, with entangled states being non-separable.

Quantum Gates and Entanglement: Unitary gates (Pauli X/Y/Z, Hadamard, rotation, CNOT) manipulate qubits while preserving normalization. Bell states demonstrate maximal entanglement. The no-cloning theorem forbids arbitrary state copying, while quantum teleportation transfers states via entanglement and classical communication.

Quantum Measurement: Measurement collapses superpositions according to Born's rule. Projective measurements and POVMs provide complementary frameworks, and the density matrix formalism extends the theory to mixed states.

Connection to Quantum Machine Learning:



These quantum computing fundamentals directly enable quantum machine learning algorithms:

- **Quantum gates** form the building blocks of parameterized quantum circuits used in variational quantum algorithms
- **Superposition** allows quantum feature maps to represent data in exponentially large Hilbert spaces
- **Entanglement** provides correlations that can express complex patterns in data
- **Measurement** extracts predictions from quantum circuits for classification and regression tasks

In subsequent units, we will build upon these foundations to implement quantum kernel methods, quantum neural networks, and hybrid classical-quantum algorithms for machine learning.

2.5 MIND MAP

[Mind map to be added]

2.6 GLOSSARY

Bell States — The four maximally entangled two-qubit states: $|\Phi_{\pm}\rangle = (1/\sqrt{2})(|00\rangle \pm |11\rangle)$ and $|\Psi_{\pm}\rangle = (1/\sqrt{2})(|01\rangle \pm |10\rangle)$.

Bloch Sphere — A geometric representation of pure single-qubit states as points on the surface of a unit sphere.

Born's Rule — The rule stating that the probability of measuring outcome i is $|\langle i|\psi\rangle|^2$, where $|\psi\rangle$ is the quantum state.

Bra — The conjugate transpose of a ket, denoted $\langle\psi|$, representing a row vector.

CNOT Gate — Controlled-NOT gate; a two-qubit gate that flips the target qubit when the control qubit is $|1\rangle$.



Computational Basis — The standard basis states $|0\rangle$ and $|1\rangle$ for a single qubit, or their tensor products for multi-qubit systems.

Density Matrix — A matrix ρ that describes a quantum state (pure or mixed), satisfying $\rho^\dagger = \rho$, $\rho \geq 0$, and $\text{Tr}(\rho) = 1$.

Dirac Notation — Bra-ket notation for quantum states, using $|\psi\rangle$ for state vectors and $\langle\psi|$ for their conjugate transposes.

Entanglement — A quantum phenomenon where the state of a multi-qubit system cannot be factored into individual qubit states.

Hadamard Gate — A single-qubit gate $H = (1/\sqrt{2})[[1,1],[1,-1]]$ that creates superposition from basis states.

Inner Product — The scalar $\langle\phi|\psi\rangle = \sum_i \phi_i^* \psi_i$, measuring the overlap between two quantum states.

Ket — A column vector representing a quantum state, denoted $|\psi\rangle$.

No-Cloning Theorem — The theorem stating that it is impossible to create an exact copy of an arbitrary unknown quantum state.

Normalization — The constraint $|\alpha|^2 + |\beta|^2 = 1$ ensuring that measurement probabilities sum to unity.

Outer Product — The matrix $|\phi\rangle\langle\psi|$ formed from a ket and a bra, creating an operator.

Pauli Gates — The three single-qubit gates X (bit flip), Y (bit and phase flip), and Z (phase flip).

POVM — Positive Operator-Valued Measure; a generalized measurement described by positive operators summing to identity.

Projective Measurement — Von Neumann measurement described by orthogonal projectors onto eigenstates.

Qubit — The fundamental unit of quantum information, a two-level quantum system that can exist in superposition of $|0\rangle$ and $|1\rangle$.



Superposition — The quantum property allowing a qubit to exist in a coherent combination of basis states.

Tensor Product — The mathematical operation $\otimes$ that combines individual qubit states into multi-qubit states.

Unitary Matrix — A matrix U satisfying $U^\dagger U = U U^\dagger = I$, representing reversible quantum operations.

2.7 SELF-ASSESSMENT QUESTIONS

Q1. Which of the following best explains why a qubit is fundamentally different from a classical bit?

- a) A qubit is faster than a classical bit
- b) A qubit can exist in a superposition of $|0\rangle$ and $|1\rangle$ simultaneously, while a classical bit must be in exactly one state
- c) A qubit uses less energy than a classical bit
- d) A qubit can store multiple classical bits within it through compression

Q2. A qubit is in the state $|\psi\rangle = (1/\sqrt{2})|0\rangle + (i/\sqrt{2})|1\rangle$. What are the probabilities of measuring $|0\rangle$ and $|1\rangle$?

- a) $P(|0\rangle) = 1/\sqrt{2}$, $P(|1\rangle) = 1/\sqrt{2}$
- b) $P(|0\rangle) = 1/2$, $P(|1\rangle) = 1/2$
- c) $P(|0\rangle) = 1/4$, $P(|1\rangle) = 3/4$
- d) $P(|0\rangle) = 0$, $P(|1\rangle) = 1$

Q3. On the Bloch sphere, the state $|+\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$ is located at the positive x-axis.

Where is the state $|-\rangle = (1/\sqrt{2})(|0\rangle - |1\rangle)$ located?

- a) North pole
- b) South pole
- c) Positive y-axis
- d) Negative x-axis



Q4. Why must all quantum gates be unitary operations?

- a) To ensure that the gates can be manufactured using standard semiconductor processes
- b) To preserve the normalization of quantum states and ensure reversibility
- c) To allow the gates to operate on multiple qubits simultaneously
- d) To ensure the gate matrices have only real-valued entries

Q5. What is the result of applying the Hadamard gate twice in succession (H^2) to a qubit in state $|0\rangle$?

- a) $|+\rangle$
- b) $|-\rangle$
- c) $|0\rangle$
- d) $|1\rangle$

Q6. A CNOT gate is applied to the two-qubit state $|10\rangle$, where the first qubit is the control and the second is the target. What is the output state?

- a) $|10\rangle$
- b) $|11\rangle$
- c) $|01\rangle$
- d) $|00\rangle$

Q7. According to Born's rule, when a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is measured in the computational basis, what determines the probability of each outcome?

- a) The real parts of α and β
- b) The squared magnitudes $|\alpha|^2$ and $|\beta|^2$
- c) The sum $\alpha + \beta$
- d) The relative phase between α and β

Q8. The Bell state $|\Phi^+\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle)$ is considered entangled because:

- a) It involves two qubits rather than one
- b) It cannot be written as a tensor product of two individual qubit states
- c) Both qubits are in a superposition state



d) Measuring one qubit takes longer than measuring a single qubit

Q9. A quantum system is described by the density matrix $\rho = (1/2)|0\rangle\langle 0| + (1/2)|1\rangle\langle 1|$.

Which of the following correctly characterizes this state?

- a) It is a pure state because it involves only basis states
- b) It is a mixed state because $\text{Tr}(\rho^2) = 1/2 < 1$, indicating a statistical mixture
- c) It is a superposition state equivalent to $(1/\sqrt{2})(|0\rangle + |1\rangle)$
- d) It is an entangled state involving two qubits

Q10. In the quantum teleportation protocol, why is classical communication between Alice and Bob necessary?

- a) To transmit the qubit's quantum state directly
- b) To share the entangled pair between Alice and Bob
- c) To convey Alice's measurement results so Bob can apply the correct corrections to recover the teleported state
- d) To verify that the teleportation was successful after it is complete

2.8 TERMINAL QUESTIONS

SHORT-ANSWER QUESTIONS

1. Explain the concept of quantum superposition and how it fundamentally differs from classical probability.
2. Describe the Bloch sphere representation of a single qubit. What states correspond to the north pole, south pole, and equatorial points?
3. What is the normalization constraint for a qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and why is it physically necessary?
4. Explain the function of the CNOT gate, including its effect on each of the four two-qubit computational basis states.



5. State the no-cloning theorem and explain why it is significant for quantum computing and quantum information.
6. Distinguish between a pure quantum state and a mixed state using the density matrix formalism. How can one determine whether a given density matrix represents a pure or mixed state?

LONG-ANSWER QUESTIONS

1. Explain the Hadamard gate in detail, including its matrix representation, its action on the computational basis states $|0\rangle$ and $|1\rangle$, and its self-inverse property. Discuss why the Hadamard gate is considered one of the most important single-qubit gates in quantum computing.
2. Describe the four Bell states, explain how they are created using a quantum circuit, and discuss why they demonstrate quantum entanglement. Include the mathematical concept of non-separability in your answer.
3. Explain the quantum measurement process in detail, covering Born's rule for calculating measurement probabilities, the concept of measurement collapse, and how measurement in different bases (computational basis versus Hadamard basis) yields different results for the same quantum state.
4. Describe the quantum teleportation protocol step by step, from the initial shared entanglement through Alice's measurement to Bob's correction operations. Explain the roles of entanglement and classical communication, and discuss why teleportation does not allow faster-than-light information transfer.

2.9 ANSWERS

2.9.1 SELF-ASSESSMENT ANSWERS

Q1. (b)



Q2. (b)

Q3. (d)

Q4. (b)

Q5. (c)

Q6. (b)

Q7. (b)

Q8. (b)

Q9. (b)

Q10. (c)

2.9.2 TERMINAL QUESTION ANSWERS

1. Quantum superposition allows a qubit to exist in a coherent combination of basis states $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where both α and β are non-zero. This differs fundamentally from classical probability. In classical probability, a coin is definitively heads or tails—we simply lack knowledge of which. In quantum superposition, the qubit genuinely exists in both states simultaneously until measured. This is evidenced by quantum interference: the amplitudes can constructively or destructively interfere, producing measurement statistics that no classical probability mixture can explain. Superposition enables quantum parallelism, where a single quantum operation processes all basis states simultaneously.
2. The Bloch sphere is a geometric representation of single-qubit pure states as points on the surface of a unit sphere. A qubit state $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$ is parameterised by polar angle $\theta \in [0, \pi]$ and azimuthal angle $\phi \in [0, 2\pi)$. The **north pole** ($\theta = 0$) corresponds to $|0\rangle$, the **south pole** ($\theta = \pi$) corresponds to $|1\rangle$, and **equatorial points** represent equal superpositions with different relative phases. For example, the positive x-axis is $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and the negative x-axis is $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. Single-qubit gates correspond to rotations on this sphere.



3. The normalization constraint requires $|\alpha|^2 + |\beta|^2 = 1$ for any qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. This is physically necessary because $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of measuring the qubit in states $|0\rangle$ and $|1\rangle$ respectively, according to Born's rule. Since the measurement must yield one of these outcomes with certainty, the probabilities must sum to unity. Violation of this constraint would mean the total probability of all outcomes is not 100%, which is physically meaningless. Unitary quantum gates automatically preserve this constraint, ensuring valid quantum states remain valid after operations.
4. The CNOT (Controlled-NOT) gate is a two-qubit gate that flips the target qubit if and only if the control qubit is $|1\rangle$. Its effect on the four computational basis states is: $|00\rangle \rightarrow |00\rangle$ (control is 0, target unchanged), $|01\rangle \rightarrow |01\rangle$ (control is 0, target unchanged), $|10\rangle \rightarrow |11\rangle$ (control is 1, target flipped), and $|11\rangle \rightarrow |10\rangle$ (control is 1, target flipped). The CNOT gate is essential for creating entanglement: applying a Hadamard gate followed by CNOT to $|00\rangle$ produces the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$. It is a universal two-qubit gate for quantum computation.
5. The no-cloning theorem states that it is impossible to create an exact copy of an arbitrary unknown quantum state. Mathematically, there exists no unitary operation U such that $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$. This follows from the linearity of quantum mechanics and can be proved by contradiction. The theorem is significant because it guarantees the security of quantum key distribution protocols (eavesdroppers cannot copy quantum states without detection), prevents trivial violation of the uncertainty principle, and distinguishes quantum information from classical information, which can be freely copied.
6. A **pure state** represents a quantum system with complete knowledge and is described by a single state vector $|\psi\rangle$ or equivalently by the density matrix $\rho = |\psi\rangle\langle\psi|$. A **mixed state** represents a statistical ensemble of pure states due to classical uncertainty and is described by $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ where p_i are classical probabilities. To distinguish



them, compute $\text{Tr}(\rho^2)$: for a pure state $\text{Tr}(\rho^2) = 1$, while for a mixed state $\text{Tr}(\rho^2) < 1$. For example, the maximally mixed state $\rho = (1/2)I$ has $\text{Tr}(\rho^2) = 1/2$, confirming it is mixed. Pure states have off-diagonal coherence terms; mixed states may lack them.

7. The Hadamard gate is one of the most fundamental single-qubit gates in quantum computing. Its matrix representation is:

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

When applied to the computational basis states: $H|0\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle) = |+\rangle$ and $H|1\rangle = (1/\sqrt{2})(|0\rangle - |1\rangle) = |-\rangle$. The Hadamard gate creates an equal superposition from a definite state, distributing probability equally between $|0\rangle$ and $|1\rangle$.

A crucial property is that H is **self-inverse**: $H^2 = I$ (the identity). Applying H twice returns the qubit to its original state: $H(H|0\rangle) = H|+\rangle = |0\rangle$. This follows from the matrix multiplication $(1/\sqrt{2})^2 \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = I$.

The Hadamard gate is indispensable because it serves as the primary tool for creating superposition, which is the starting point for most quantum algorithms. In algorithms like Grover's search and the Quantum Fourier Transform, Hadamard gates initialise qubits into uniform superpositions. On the Bloch sphere, H performs a rotation of π radians about an axis halfway between x and z , swapping the computational and Hadamard bases. Combined with phase gates and CNOT, the Hadamard gate forms a universal gate set capable of implementing any quantum computation.

8. The four Bell states are maximally entangled two-qubit states:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned}$$

They are created using a simple circuit: apply a Hadamard gate to the first qubit, then a CNOT gate with the first qubit as control and the second as target. Starting from $|00\rangle$: H



on qubit 1 gives $(1/\sqrt{2})(|0\rangle + |1\rangle)|0\rangle$, then CNOT produces $(1/\sqrt{2})(|00\rangle + |11\rangle) = |\Phi^+\rangle$. The other Bell states are obtained by starting from $|01\rangle$, $|10\rangle$, or $|11\rangle$.

Bell states demonstrate entanglement because they are **non-separable**—they cannot be written as a tensor product of individual qubit states. For $|\Phi^+\rangle$, there exist no single-qubit states $|a\rangle$ and $|b\rangle$ such that $|\Phi^+\rangle = |a\rangle \otimes |b\rangle$. This can be verified by attempting the factorisation and arriving at a contradiction. Physically, measuring one qubit instantaneously determines the other's state regardless of distance. If qubit A is measured as $|0\rangle$, qubit B must be $|0\rangle$ (and similarly for $|1\rangle$). These correlations are stronger than any classical correlation, as demonstrated by violations of Bell's inequalities.

9. Quantum measurement extracts classical information from a quantum system, fundamentally altering the state in the process.

Born's rule provides the probability of each measurement outcome. For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ measured in the computational basis, the probability of obtaining $|0\rangle$ is $P(0) = |\alpha|^2$ and the probability of $|1\rangle$ is $P(1) = |\beta|^2$. For example, the state $(1/\sqrt{3})|0\rangle + \sqrt{2/3}|1\rangle$ yields $P(0) = 1/3$ and $P(1) = 2/3$.

Measurement collapse means that after measurement, the qubit is projected into the measured state. If we obtain outcome $|0\rangle$, the post-measurement state is $|0\rangle$ regardless of the original superposition. This is irreversible—the original amplitudes α and β are lost. Repeated measurements of the same state will always return the same result.

Measurement in different bases yields different statistics. The state $|0\rangle$ measured in the computational basis always gives $|0\rangle$ with certainty. However, measured in the Hadamard basis $\{|+\rangle, |-\rangle\}$, we express $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$, giving equal probability for each outcome. Similarly, the state $|+\rangle$ is certain in the Hadamard basis but random in the computational basis. The choice of measurement basis determines which information is extracted, reflecting the complementarity principle—no single measurement reveals complete information about the quantum state.



10. Quantum teleportation transfers a quantum state from one location (Alice) to another (Bob) using entanglement and classical communication.

Setup: Alice and Bob share a Bell pair $|\Phi^+\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle)$. Alice has an additional qubit in unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that she wishes to teleport.

Step 1 — Alice's operations: Alice applies a CNOT gate with her unknown qubit as control and her Bell pair qubit as target, then applies a Hadamard gate to her unknown qubit. The three-qubit system is now in a superposition of four terms, each correlated with a specific two-bit measurement outcome.

Step 2 — Alice's measurement: Alice measures both her qubits in the computational basis, obtaining one of four outcomes: 00, 01, 10, or 11. This measurement collapses the state, leaving Bob's qubit in a specific state that depends on Alice's result.

Step 3 — Classical communication: Alice sends her two classical measurement bits to Bob through a classical channel.

Step 4 — Bob's correction: Based on Alice's bits, Bob applies correction operations: 00 $\rightarrow$ no operation, 01 $\rightarrow$ X gate, 10 $\rightarrow$ Z gate, 11 $\rightarrow$ ZX gates. After correction, Bob's qubit is in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

Entanglement provides the quantum channel connecting Alice and Bob, but classical communication is essential—without it, Bob's qubit appears as a random mixed state. This classical communication requirement ensures that **no faster-than-light information transfer** occurs: the quantum correlations alone cannot transmit information, as Bob's local measurement statistics are uniformly random until he applies the corrections based on Alice's classical message.

2.10 REFERENCES

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